Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-Powered DNA Digital Twin for Precision Polypharmacy Management System for Elderly Healthcare Crisis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Information & Systems	Other Medical Information & Systems	Therapeutic Devices & Diagnostic Devices	Intelligent interpretation Systems
Concept of Proposed Technology	**1. Project Overview and Existing Infrastructure** This initiative seeks to validate a verified Genomic Digital Twin Platform, currently at Technology Readiness Level (TRL) 6, through the integration of advanced pharmacological and consultative modules. The foundational platform serves as a longitudinal data management system that transforms Whole Genome Sequencing (WGS) data into a searchable genomic blueprint. By synthesizing data from multiple international reference databases, the system provides comprehensive genetic risk profiling across 21 physiological organ systems. The underlying architecture utilizes an Agentic AI framework consisting of numerous domain-specific autonomous agents. Unlike traditional monolithic large language models, this multi-agent system employs a consensus-based negotiation mechanism to synthesize clinical recommendations based on specialized domain expertise. To date, the platform has completed pilot study and deployment at multiple clinical sites. **2. Technical Enhancement Modules** The three-year research and development plan focuses on the implementation of three strategic modules designed to improve clinical decision support and data utility. The first component is a Drug-Drug Interaction (DDI) screening engine. This module utilizes real-time API integration with national health insurance and drug safety review systems to provide automated medication alerts. Technical objectives include scaling the interaction database from an initial 5,000 drug pairs to over 8,000 pairs			

	by the third year, ensuring high-fidelity, personalized safety notifications. Secondly, in coordination with DDI enhancements, the project will implement a Gene-Drug Interaction (GDI) analysis engine. This system matches patient-specific genotypes against established international clinical guidelines to provide evidence-based prescribing support. The project aims to expand the scope of this analysis from 450 gene-drug pairs in the initial phase to 1,000 pairs by the final phase of development. Thirdly, it requires validation of the accuracy. Consequently, the AI-generated reports must undergo rigorous and systematic validation to ensure that all clinically actionable claims—especially those concerning drug efficacy, adverse effects, and contraindications—are faithfully grounded in and consistent with the underlying peer-reviewed evidence. By deploying these techniques as lightweight wrapper layers around the AI-Powered DNA Digital Twin for Precision Polypharmacy Management System's existing RAG and LLM pipelines, the collaboration enhances model reliability without introducing significant computational or operational overhead. This partnership structure enables each party to concentrate on its respective strengths, maximizing both efficiency and technical impact. **3. Consultation Interface and System Integration** To facilitate user interaction and data interpretation, the platform will integrate a Large Language Model (LLM)-powered consultation interface. This system will be deployed across five specialized domains: Medication Safety, Disease Risk, Diet/Nutrition, Exercise/Wellness, and Clinical Results Interpretation. The interface is required to provide citation-verified, bilingual support in both the local and English languages. The overall system architecture is based on a loosely coupled, API-driven design. Under this framework, the consultation interfaces function as an interpretive layer that presents findings generated by the core genomic engine without duplicating analytical workloads. This modular approach ensures computational efficiency and allows for scalable deployment across different segments of the platform while maintaining a unified data source for genomic and pharmacological analysis. **4. Enhancement of Evaluation and Characterization of GDIs in Clinical Pharmacotherapy** GDIs arise from the combined influence of pharmacogenomic variation and concomitant drug therapy on drug-metabolizing enzymes and transporters, often resulting in clinically meaningful alterations in drug exposure, efficacy, and safety. It seeks to systematically characterize GDIs across these principal categories—inhibitory interactions, induction interactions, and phenoconversion interactions—using mechanistic frameworks and clinically relevant examples. It focuses initially on metabolizing enzymes and subsequently extend to drug transporters, with the objective of improving prediction, interpretation, and clinical management of GDIs.
Rationale for Joint R&D Project	**1. Public Health Context and Technical Necessity** By 2030, the domestic elderly population is projected to reach 26.7%, with senior-related medical expenditures exceeding ₩44 trillion. A significant clinical challenge within this demographic is polypharmacy; approximately 35.4% of elderly patients are prescribed five or more medications concurrently. Data indicates that 53.7% of this population utilizes Potentially Inappropriate Medications (PIMs), which are associated with a 25% increase in mortality risk. While existing national Drug Utilization Review (DUR) systems successfully identify Drug-Drug Interactions (DDI), they do not currently account for Gene-Drug Interactions (GDI). Given that genetic polymorphisms can account for 20% to 95% of variability in drug response, there is a critical requirement for a localized, data-driven system that integrates pharmacogenomics into standard clinical decision support.

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2. Strategic R&D Collaboration Framework

The successful development of an pharmacogenomics platform requires validation of the accuracy of language model output. The proposed collaboration brings together specialized capabilities that cannot be achieved independently:

- **Technological Lead:** Contributes proprietary Agentic AI architecture and a comprehensive database of approximately 348,000 regional clinical cases. This addresses the historical 95% Caucasian bias in pharmacogenomics literature. The technological lead provides the necessary infrastructure for real-time national insurance API integration and established partnerships with local medical institutions for rapid validation.
- **Validation Lead:** The pharmacogenetics-specialized, DNA-based digital twin generates personalized clinical reports by leveraging a language model that synthesizes information from reliable, peer-reviewed literature and curated medical databases. By integrating individual genomic profiles with established pharmacogenomic knowledge, the system produces patient-specific insights, including drug response predictions and interaction risks. While this approach enables scalable and highly personalized clinical reporting, the generative nature of the language model introduces uncertainty regarding how precisely the synthesized outputs reflect the underlying evidence base. Even when source materials are authoritative, intermediate reasoning steps, summarization, and cross-document synthesis can lead to subtle distortions, omissions, or overgeneralizations that are difficult to detect in downstream clinical use.

  As a result, the generated reports require systematic validation to ensure that each clinically actionable statement—particularly those related to drug efficacy, adverse reactions, and contraindications—can be traced back to and is consistent with the original peer-reviewed sources. Validation is especially critical in pharmacogenetics, where recommendations may differ across populations and where errors can directly impact patient safety. Establishing formal validation mechanisms will transform the digital twin from a powerful generative system into a clinically trustworthy decision-support tool by quantifying confidence, certifying evidence alignment, and clearly communicating the reliability of AI-generated recommendations to clinicians.

3. Synergy and Technical Rationale

The partnership is structured to leverage the comparative advantages. The partnership is structured to leverage the complementary strengths of both teams across two critical dimensions: operational efficiency and technical rigor in trustworthy AI. The AI-Powered DNA Digital Twin for Precision Polypharmacy Management System contributes a mature, production-grade AI ecosystem, including a proprietary Agentic AI architecture. This existing infrastructure enables rapid deployment, continuous data ingestion, and real-world evaluation of validation methods without disrupting clinical workflows, significantly reducing development and translation timelines. In parallel, Dr. Sulam's team provides state-of-the-art expertise in trustworthy machine learning, including theoretically grounded validation, calibration, and certification techniques for LLM-generated outputs. By integrating these methods as lightweight wrapper layers around the AI-Powered DNA Digital Twin for Precision Polypharmacy Management System's existing RAG and LLM pipelines, collaboration achieves high operational efficiency while substantially improving model reliability. This division of the partnerships allows each partner to focus on its comparative advantage.

4. Requirement for Joint Implementation

The contribution that Dr. Sulam's team at Hopkins could make to this project is to adapt and provide methods to improve the trustworthiness of the AI systems that this AI-Powered DNA Digital Twin for Precision Polypharmacy Management System is currently using to provide reports, validating and certifying degrees of uncertainty in the output generated by your LLMs. While the current software generates diagnostic reports and drug interaction assessments using RAG and LLMs trained on

	trusted medical databases (textbooks, PubMed), there is currently no systematic method to verify how accurately these AI-generated outputs reflect the original source material—a "black box" problem that undermines clinical confidence. Dr. Sulam's lab specializes in trustworthy machine learning and has developed methods with theoretical guarantees that can certify accuracy levels of predictive models, calibrate outputs to ensure specified portions are verifiably correct, and provide wrapper algorithms that validate AI-generated content against authoritative sources. The collaboration will develop validation methods to quantify the trustworthiness of the AI-Powered DNA Digital Twin for Precision Polypharmacy Management System reports, particularly for critical clinical recommendations like drug interaction warnings, thereby providing clinicians with confidence levels for AI-generated recommendations and ensuring patient safety in decision-making scenarios.
Key Words	1. Language Model Accuracy Validation 1. DNA Based Digital Twin 2. Pharmacogenomics 3. Agentic AI 4. Polypharmacy Management 5. Drug Interaction Analysis

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (6)	End of the Project	Level (8)
R&D Objectives	Validated the AI-Powered DNA Digital Twin for Precision Polypharmacy Management System: Addressing Elderly Healthcare Crisis, achieving >60% reduction in Adverse Drug Reactions (ADRs) for elderly patients (age >65) taking >5 medications while maintening >90% textual accuracy.			

R&D Target Specifications

Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters

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Performances	Target
Patent Filings	3 patents filed
Hospital Deployment	2-3 hospitals (commercial contracts)
Revenue	Year 3 cumulative >₩290M
Test Volume	Year 3 cumulative >1,950 tests
SCI(E) Publications	2-3 papers published
DDI Detection Sensitivity	>95%
GDI Prediction Accuracy	>90%
AI Chatbot Textual Accuracy	>90%
System Response Time	<2 seconds
Healthcare Cost Savings (Modeling)	>₩50M/patient/year
Regulatory Pathway Assessment	Requirements analysis completed

R&D Plan

36-Month R&D Roadmap (TRL 6 to 8)

This project transitions an AI-driven clinical decision support system (CDSS) from computational validation to large-scale commercial deployment.

Year 1: Foundation and Computational Validation (TRL 6 to 7)

- **Focus:** Data integration, AI model enhancement, and retrospective validation.
- **Key Tasks:** Integrate HIRA/MFDS APIs (5,000+ DDI/GDI pairs) and build a unified knowledge graph. Fine-tune LLMs for medical terminology and implement a 21-agent consensus algorithm.
- **Milestones:** Diagnostic accuracy > 98.8%; first patent filed; first SCI(E) paper published; Chatbot prototype completed.
- **Research Institute Role:** Provide multi-ethnic case data for independent validation and co-author manuscripts.

Year 2: Textual Validation (TRL 7 to 8)

- **Focus:** Calibrate outputs to ensure specified portions are verifiably correct, and provide wrapper algorithms that validate AI-generated content against authoritative sources.
- **Key Tasks:** Define verifiable output units, wrapper algorithm development, integration and

	system optimization. • **Milestones:** Formal definition of verifiable report components and validation criteria, functional wrapper capable of claim-level source verification and confidence annotation, end-to-end integration within the digital twin system and pilot testing in a controlled clinical setting, and comprehensive evaluation report demonstrating improved trustworthiness and operational feasibility. • **Research Institute Role:** methodological leadership, algorithm development, evaluation and verification, and knowledge transfer.. **Year 3: Commercial Deployment and Regulatory Preparation (TRL 8)** • **Focus:** Hospital-wide activation, revenue generation, and regulatory certification. • **Key Tasks:** Deploy real-time CDSS dashboards and patient-facing chatbots in 3+ hospitals. Monitor alert override rates (target < 20%) and validate chatbot accuracy. • **Milestones:** Cumulative revenue > 290M KRW; 1,950+ tests completed; final patent filed; regulatory pathway assessment for MFDS Class 2 finalized. • **Research Institute Role:** Strategic advisory for US FDA 510(k) pathway and global market entry roadmap.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Technical Contributions and Research Scope** • Focus on validation and calibration of safety-critical outputs only, prioritizing drug–drug and drug–gene interaction statements generated by LLM components. • Develop lightweight claim-level confidence calibration methods for LLM-generated recommendations. • Implement post-generation wrapper checks to verify alignment with authoritative pharmacogenomics sources. **Resources and Infrastructure** • Adapt and apply existing trustworthy ML methodologies and evaluation pipelines without developing new model architectures. • Use limited computational resources for prototyping and targeted validation experiments. • Deliver modular, easily deployable validation components compatible with the existing RAG/LLM system. **Primary Tasks and Deliverables** • Identify a small set of verifiable clinical claims within generated reports. • Produce a prototype calibration module assigning confidence scores to the reported claims. • Deliver a basic validation wrapper that checks citation consistency and evidence alignment. • Provide concise technical documentation and a short evaluation summary demonstrating feasibility and impact.

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Expected R&R of Korean SME Partner	**Expected R&R of Korean SME Partner** **Technical Contributions:** Lead platform development (DDI/GDI engines, chatbot system), EMR integration (HL7 FHIR for 5 hospital systems), HIRA/MFDS API management (authentication, real-time synchronization) **Development Tasks:** Train AI models (400,000+ cases), fine-tune LLM (Korean-English medical terminology), implement 21-agent Agentic AI orchestration, build RAG pipeline for citation-verified responses **Testing/Validation:** Execute 347,736-case retrospective validation, manage 10,000-patient RWE collection, conduct 500-patient RCT, usability testing (SUS >80) **Resource Commitments:** Five hospital partnership agreements (Seoul National University Hospital, Asan Medical Center, Samsung Medical Center, Severance Hospital, Catholic Medical Center), IRB submissions, patient recruitment, ISO 27001 data security **Collaboration:** Facilitate quarterly workshops with Johns Hopkins, semi-annual TIPA progress reports, DSMB coordination **Commercialization:** MFDS Class 2 application, HIRA reimbursement negotiation, ISO 13485 certification, commercial launch (10 hospitals, 100 physician training), marketing strategy, three patent filings

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technological Achievements** • World's first DNA Digital Twin integrating DDI/GDI/Agentic AI for polypharmacy management • Eight patents (5 existing + 3 new: Agentic AI architecture, GDI prediction algorithm, CDSS workflow) • Two to three peer-reviewed publications in high-impact journals (IF >10) **Clinical Impact (100,000 Elderly Patients)** • 60% ADR reduction: 15,000 ADRs prevented annually • 700 lives saved annually (25% mortality risk reduction from PIM elimination) • 30-day hospital readmission reduction: 25% • Medication adherence improvement: 85% **Economic Impact** • ₩50 million savings per patient per year (drug costs + hospitalization + ADR treatment) • National healthcare savings: ₩5 trillion annually (100,000 patients) • Return on Investment (ROI): 1:30 (genetic test cost vs. 10-year savings) **Commercialization Potential** • **Korea:** 10 hospitals (2027) → 50 hospitals (2028) → national rollout (2029) • **United States:** Johns Hopkins endorsement facilitates Medicare coverage negotiations (2028), partnerships with CVS Health, UnitedHealth • **Middle East/EU:** Export partnerships leveraging CES 2026 & Edison Award 2026 recognition (2029) • **Revenue Projections:** ₩50 billion (2028), ₩200 billion (2030) **Regulatory Milestones** • MFDS Class 2 Medical Device Software approval (Year 3 Q3) • HIRA reimbursement code for pharmacogenomic panel testing (Year 3 Q4) • ISO 13485 certification (Year 3 Q4) • FDA 510(k) pathway initiated with Johns Hopkins (Year 3 Q4) **Broader Industrial Impact**

	• Establish Korea as global leader in precision medicine AI platforms • Create export opportunities for Korean healthcare AI technologies (estimated $2 billion global market by 2030) • Enable integration with national health insurance digital transformation initiatives • Provide framework for expanding platform to other medication classes (oncology, immunotherapy, antibiotics)

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